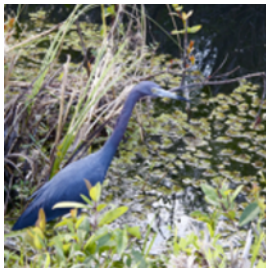


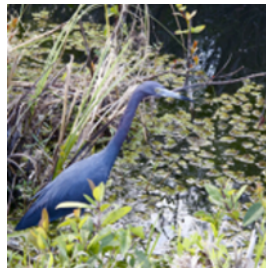
Adversarial robustness: a tiny worst-case change flips the label



clean: “little blue heron” ✓



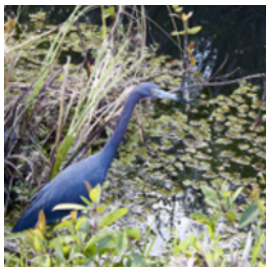
perturbation ($\times 10$)



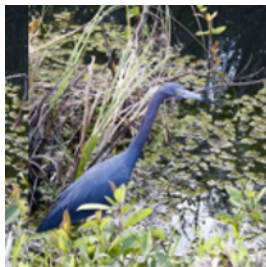
adversarial: “tusker” ✗

The two images differ by an ℓ_∞ change of $1/255$, invisible to a human; the label flips.

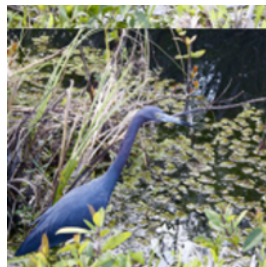
Invariance: shifting the image keeps the label



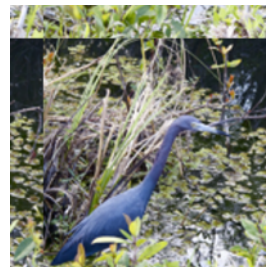
“little blue heron” ✓
original



“little blue heron” ✓
shift (20,0)



“little blue heron” ✓
shift (0,20)



“little blue heron” ✓
shift (28,28)